

Bandon Public Transport Accessibility Analysis

Bandon, County Cork, Republic of Ireland

SEL-002 | Public Transport Accessibility & Vulnerability

Prepared as a reproducible GIS-based spatial evidence study



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Abstract

Bandon is one of the fastest-growing towns in West Cork and occupies a distinctive position within the regional settlement hierarchy. According to Census 2022, the town has a resident population of 8,196 and functions simultaneously as a commuter town within the Greater Cork metropolitan area and as a gateway to the wider West Cork region. Its strategic location approximately 30 km south-west of Cork City makes it an important origin and destination for daily commuting, education, healthcare, retail, and regional services.

This dual role raises critical questions about the performance of the public transport network. While Bandon is connected to Cork City and surrounding settlements through regional bus services, the extent to which those services provide an effective and equitable alternative to private-car travel varies across the town. Accessibility is therefore considered not only in terms of the physical location of bus stops, but also in relation to the characteristics and needs of the population living within different parts of Bandon.

Research Questions

Is public transport a viable alternative to private-car commuting between Bandon and Cork City?

Which areas are potentially most vulnerable to limited public transport accessibility because of their demographic, health, employment, or household characteristics?

The first question focuses on the spatial performance of the public transport system through walking accessibility and service frequency. The second recognises that accessibility should be evaluated within a broader social context, acknowledging that some population groups are more likely to depend on public transport than others. A bus stop may be geographically accessible, yet still fail to provide meaningful accessibility if services are infrequent or if populations with greater mobility needs, remain poorly served.

Executive Summary

The hotspot analysis demonstrates that public transport accessibility in Bandon is spatially uneven and strongly centralised. Areas surrounding the town centre benefit from the highest levels of accessibility due to their proximity to the four principal bus stops and the concentration of scheduled bus services. In contrast, several peripheral residential neighbourhoods experience substantially lower accessibility while simultaneously containing populations with characteristics associated with greater transport vulnerability.

The findings indicate that accessibility and vulnerability do not follow the same spatial pattern. Some neighbourhoods have relatively low accessibility despite comparatively low vulnerability, while others combine limited accessibility with demographic and socio-economic characteristics that increase potential dependence on public transport. These latter locations emerge as Critical and High Priority Accessibility-Vulnerability Hotspots.

The analysis provides a spatial evidence base for identifying areas where improvements to walking infrastructure, bus stop provision, service frequency, or broader transport planning interventions could deliver the greatest social benefit.

Answers to the Research Questions

Is public transport a viable alternative to private-car commuting between Bandon and Cork City?

Answer: Public transport is more viable for residents within the central accessibility corridor, where walking access to principal stops is strongest and peak-period services are more available. Viability becomes more constrained across several peripheral residential neighbourhoods where network walking journeys are longer and service coverage is weaker. The analysis therefore supports a spatially differentiated conclusion rather than a single town-wide assessment.

Which areas of Bandon are potentially most vulnerable to limited public transport accessibility?

Answer: The Critical and High Priority classifications occur primarily in outer residential areas where lower relative accessibility overlaps with higher population vulnerability. The vulnerability model captures older age, health and disability characteristics, economic-status indicators and households without a car. These areas represent locations where transport disadvantage may have greater consequences and where more detailed planning assessment is warranted.

1. Understanding Public Transport Accessibility

Public transport accessibility describes the ease with which people can reach and use public transport and, through it, access destinations and opportunities. In this study, accessibility is treated as a spatial and multidimensional concept rather than a single distance measure.

1.1 Public Transport Accessibility in Urban and Regional Planning

Accessibility is concerned with the relationship between people, transport systems and opportunities. A location may be physically close to a bus stop but still have relatively weak effective accessibility if services are infrequent. Conversely, a location with a somewhat longer walk may have stronger effective accessibility where frequent services are available. The Bandon analysis therefore separates walking access from service availability before combining them in the composite analysis.

Opportunities: Accessibility links transport provision to access to employment, education, healthcare, services and other opportunities.

Mobility vs. Accessibility: Mobility concerns movement itself, whereas accessibility concerns the opportunities that movement enables.

Quality of Life: Accessibility is relevant to quality of life because transport conditions influence people's ability to participate in social and economic life.

Equity: Transport equity requires consideration of both the distribution of service provision and the distribution of population need.

1.2 Accessibility as a Spatial Concept

The Bandon framework calculates accessibility through several related spatial dimensions: distance, time, opportunity, network connectivity and service availability. Walking distance and network travel time describe the physical effort required to reach a stop, while GTFS-derived service frequency introduces a temporal dimension.

Distance: How far a residential area is from a public transport stop through the pedestrian network.

Time: The estimated walking time required to reach the transport network.

Opportunity: The potential to use public transport once a stop is reached.

Network Connectivity: The pedestrian routes available for reaching stops rather than relying solely on straight-line distance.

Service Availability: The frequency and temporal intensity of public transport services derived from GTFS feeds.

1.3 Accessibility in Small Towns

Smaller commuter towns can display a sharp transition between a relatively well-served urban core and lower-density peripheral residential areas. Bandon provides a useful case study because the mapped results show that accessibility is spatially structured rather than uniform across the study boundary.

1.4 Accessibility and Social Equity

Transport disadvantage becomes particularly important where weaker accessibility overlaps with population characteristics associated with greater potential dependence on public transport. The vulnerability layer is therefore not

treated as a measure of transport accessibility itself. It is a separate population-need dimension that is combined with accessibility at the hotspot stage.

1.5 Research Context

The Bandon study forms part of the Spatial Evidence Lab portfolio and applies a reproducible GIS approach to a small-town transport question.

Accessibility → Mobility → Opportunity → Quality of Life

2. Study Area and Analytical Framework

The purpose of the study is to assess how public transport accessibility varies spatially across Bandon and to identify locations where lower accessibility coincides with greater population vulnerability.

2.1 Study Area Overview

The study area is defined by the mapped study boundary around Bandon and its surrounding residential and settlement areas. The context map identifies public transport stops and the local road network. The mapped extent captures the town centre and peripheral areas including Gully, Cloughmacsimon, Kilbrogan, Ballylanglely, Currough, and Knocknagrohan.

The study uses Small Areas as the principal reporting geography for population vulnerability and spatial comparison. Public transport stops are represented as point features, while the road and pedestrian network provides the spatial context for network-based access analysis.

2.2 Study Context and Population Vulnerability Assessment

Bandon has a resident population of 8,196 (Census 2022) and functions as a key regional settlement and commuter node. Bandon was selected because its defined urban core and surrounding lower-density residential peripheries make it an ideal small-town case study for examining the transition between high and lower public transport accessibility.

Population Vulnerability as a Complementary Dimension

To evaluate transport equity, the analysis incorporates Census 2022 Small Area data to construct a Population Vulnerability Index. This index does not measure accessibility itself; instead, it identifies characteristics that may increase potential dependence on public transport or indicate greater vulnerability to transport disadvantage.

Census Variable	Description	Relevance to Vulnerability Assessment
T1_1AGE70_74T	Population aged 70-74	Identifies older residents who may have increased dependence on accessible transport.
T1_1AGE75_79T	Population aged 75-79	Represents populations with potentially greater mobility needs.
T1_1AGE80_84T	Population aged 80-84	Captures areas with higher concentrations of elderly residents.
T1_1AGEGE_85T	Population aged 85+	Represents population most likely to require highly accessible services.
T1_1AGETT	Total population	Denominator for population-based vulnerability measures.
T4_5RP / T8_1_RT	Retired persons	Represents populations less likely to rely on work travel but dependent on local transport.
T8_1_LTUT	Long-term unemployed	Represents socio-economic vulnerability and transport disadvantage.
T8_1_UTWSDT	Permanent sickness/disability	Represents populations experiencing greater mobility constraints.
T12_1_T	Persons with a disability	Measures populations requiring accessible transport infrastructure.
T12_3_BT	Bad general health	Indicates communities with health-related transport needs.
T12_3_VBT	Very bad general health	Identifies areas with high health-related transport vulnerability.
T15_1_NC	Households without a car	Households potentially most dependent on public transport.
T11_1_BUT	Travelling by bus/minibus	Provides evidence of existing public transport use.
T11_2_T1-T8	Departure time for work/school	Captures commuting demand during morning departure periods.
T11_2_NS / T11_2_T	Leaving time not stated / Total	Denominator for analysing commuting time distributions.

Dimensions of Population Vulnerability

These Census variables are grouped into four core vulnerability domains:

Demographic Vulnerability: Older population (70+ years), retired persons.

Health and Mobility Vulnerability: Disability, permanent sickness, bad and very bad general health.

Socio-Economic Vulnerability: Long-term unemployment and retirement-related indicators.

Transport Dependence: Households without a motor car, bus commuters, and morning departure-time patterns.

Integration Framework

The analysis strictly separates the spatial performance of the transport network from population demand and vulnerability:

Census Variable	Purpose in the Study
Public Transport Supply & Accessibility	Measured via bus stop locations, walking-network catchments, and peak service frequency
Observed Travel Behaviour	Represented by Census commuting patterns and bus-use variable
Population Vulnerability	Represented by age, retirement, unemployment, disability, and health indicators
Private Vehicle Availability	Represented by households without access to a motor car

2.3 Research Questions and Objectives

Primary Research Question: How can GIS-based accessibility and vulnerability analysis identify areas of transport disadvantage in Bandon?

- Research Objectives:
- Map the spatial coverage of public transport stops using walking-network analysis.
 - Measure walking-distance accessibility to public transport stops.
 - Assess peak-time service availability using GTFS-derived service frequency.
 - Construct a population vulnerability layer using selected Census 2022 indicators.
 - Combine accessibility and vulnerability to identify spatial hotspots and priority areas.

2.4 Analytical Framework and Map Series

Map	Title	Subtitle
01.00	Base Context & Transit Infrastructure	Bandon Built-Up Areas (BUA) and Network Baseline
01.01	Physical Pedestrian Catchments	Network-Based Isochronal Walkability Analysis
01.02	First/Last-Mile Infrastructure Quality	Footpath Coverage and Pedestrian Crossing Networks
01.03	Headway & Operational Frequency	Peak-Period Transit Service Quality
01.04	Demographic Vulnerability & Demand	Spatial Distribution of Priority Transit-Dependent Populations
01.05	Composite Accessibility Index (CAI)	Combined Pedestrian Accessibility and Population Vulnerability
01.06	Targeted Intervention Framework	Spatial Decision-Support Priorities for Capital Investment

3. Dimensions and Methodology

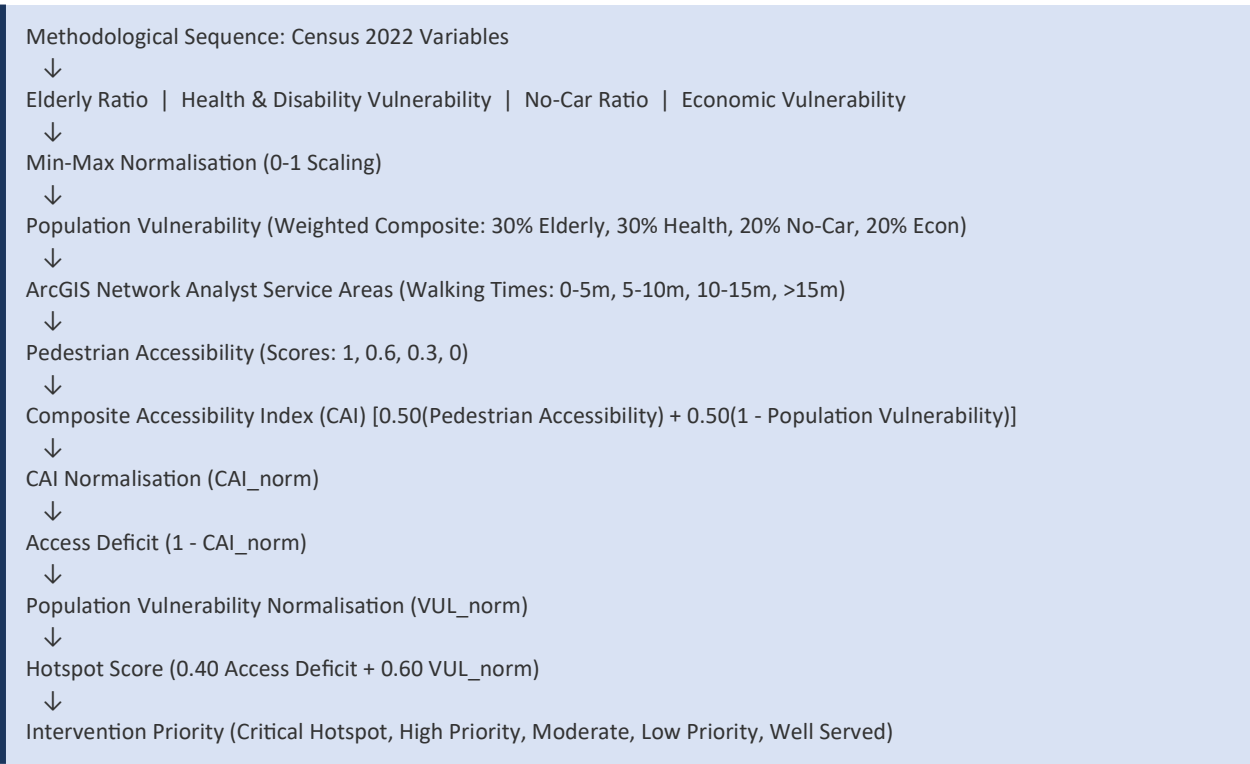
3.1 Indicators Terminology and Nomenclature

To ensure consistency throughout the report, GIS databases, cartographic products, and web dashboards, the standardized terminology presented in Table 1 must be strictly adhered to across all analytical workflows.

Concept / Measure	Standardised Terminology	Application Scope & Context
Original Census variables	indicator	Original Census-derived demographic and socio-economic variables.
Min-Max transformed variables	normalised indicator	0-1 rescaled individual census variables.
Weighted vulnerability composite	Population Vulnerability	Integrated demographic, health, car, and economic vulnerability index.
Network walking access score	Pedestrian Accessibility	ArcGIS Network Analyst decay score.
Combined supply score	Composite Accessibility Index (CAI)	50:50 combination of Pedestrian Accessibility & service availability.
Inverse CAI measure	Access Deficit	1 - Normalised CAI (measures relative accessibility gap).
Continuous spatial priority	Hotspot Score	40% Access Deficit + 60% Normalised Population Vulnerability.
Planning classification	Intervention Priority	Five-tier planning classification for targeted policy interventions.

3.2. Analytical Hierarchy & Workflow Sequence

The analytical pipeline progresses through a structured spatial workflow from raw Census 2022 variables and network service areas to a continuous Hotspot Score and categorical planning classification:



3.3 Indicator Definitions & Methodological Formulations

This section establishes the formal mathematical definitions, data dictionary formulas, methodological notes, and operational interpretations for all thirteen core indicators in the analytical framework.

Elderly Ratio

Definition: The proportion of the total population aged 70 years and over within each Small Area.

Formula: Elderly Ratio = (Age 70-74 + Age 75-79 + Age 80-84 + Age 85+) / Total Population

Interpretation: Higher values indicate a greater concentration of older residents and therefore potentially greater dependence on accessible public transport.

Health and Disability Vulnerability

Definition: A composite indicator representing the proportion of the Small Area population associated with the Census health/disability variables included in the analytical model.

Formula (Dataset Fields): Health Disability = (T12_1_T + T12_3_BT + T12_3_VBT) / Total Population

Interpretation: Higher values indicate greater potential vulnerability associated with health and disability characteristics.

Methodological Note: This should be described as a health/disability composite measure, rather than as a direct measure of the prevalence of a single health condition. (Fields: T12_1_T = Disability, T12_3_BT = Bad Health, T12_3_VBT = Very Bad Health).

No-Car Ratio

Definition: The proportion of the population associated with households recorded as having no car, used as an indicator of potential dependence on public transport.

Formula (Dataset Fields): NoCar Ratio = T15_1_NC / Total Population

Interpretation: Higher values indicate greater potential dependence on public transport because of limited household car availability.

Methodological Note: The source numerator (T15_1_NC) represents households whereas the denominator is total population. This should be explicitly documented as a model-derived proxy, rather than presented as a conventional population percentage.

Economic Vulnerability

Definition: A composite measure representing economic vulnerability using the Census economic-status variables included in the model.

Formula (Dataset Fields): Econ Vulnerability = (T8_1_RT + T8_1_UTWSDT) / Total Population

Interpretation: Higher values indicate greater concentration of the economic-status characteristics represented by the selected Census variables.

Methodological Note: The exact CSO definitions of T8_1_RT (Retired) and T8_1_UTWSDT (Unable to work due to sickness/disability) should be retained in the metadata/data dictionary. The supplied analytical table supports the formula above but does not, by itself, establish alternative labels for these fields.

Normalised Vulnerability Indicators

Definition: Each of the four vulnerability indicators is transformed to a common 0-1 scale using min-max normalisation.

Formula: $X_{\text{norm}} = (X - X_{\text{min}}) / (X_{\text{max}} - X_{\text{min}})$
where X = Small Area indicator value, X_{min} = minimum across analytical population, X_{max} = maximum across analytical population.

Interpretation: 0 = lowest observed vulnerability; 1 = highest observed vulnerability. The four resulting normalised measures are:

- Elderly_Norm
- Health_Norm
- NoCar_Norm
- Econ_Norm

The normalisation provides a common scale for combining indicators measured using different units and distributions.

Population Vulnerability

Definition: A weighted composite measure representing the relative vulnerability of the Small Area population to potential public transport exclusion.

Formula: $\text{Population Vulnerability} = 0.30(\text{Elderly_Norm}) + 0.30(\text{Health_Norm}) + 0.20(\text{NoCar_Norm}) + 0.20(\text{Econ_Norm})$

Component Domain	Assigned Weight (%)
Elderly population (Elderly_Norm)	30%
Health and disability (Health_Norm)	30%
No-car households (NoCar_Norm)	20%
Economic vulnerability (Econ_Norm)	20%
Total Composite Weight	100%

Interpretation: Higher values indicate greater relative population vulnerability.

Pedestrian Accessibility

Definition: A network-based measure of walking accessibility from Small Areas to principal public transport stops, derived using ArcGIS Network Analyst service-area analysis.

The analysis uses a standard walking speed of 5 km/h and four network walking-time bands:

Walking Time Band	Accessibility Score	Approximate Distance at 5 km/h
0 - 5 minutes	1.0	Up to ~417 m
5 - 10 minutes	0.6	~417 m - 833 m
10 - 15 minutes	0.3	~833 m - 1,250 m
> 15 minutes	0.0	Greater than ~1,250 m

Interpretation: Higher scores represent better pedestrian access to public transport. A score of 1.0 represents the strongest accessibility class, while 0.0 represents locations outside the 15-minute network walking threshold. The measure is based on true network walking time, rather than straight-line distance. Note: The terms accessibility score or network accessibility score are preferable to describing the 1.0, 0.6, 0.3 and 0.0 pedestrian values as percentages.

Composite Accessibility Index (CAI)

Definition: A combined measure of physical pedestrian accessibility and population vulnerability.

Formula: $\text{CAI} = 0.50(\text{Pedestrian Accessibility}) + 0.50(1 - \text{Population Vulnerability})$

The inverse vulnerability term (1 - Population Vulnerability) is used because higher vulnerability represents poorer conditions from an accessibility perspective.

Interpretation: Higher CAI indicates better accessibility relative to population vulnerability; lower CAI indicates poorer accessibility and/or higher population vulnerability. The two dimensions contribute equally (50:50) to the initial Composite Accessibility Index.

CAI Normalised (CAI_norm)

Definition: The Composite Accessibility Index transformed to a 0-1 scale using min-max normalisation.

Formula: $CAI_norm = (CAI - CAI_min) / (CAI_max - CAI_min)$

Interpretation: 0 = lowest relative accessibility; 1 = highest relative accessibility. This measure provides a consistent basis for calculating the access deficit.

Population Vulnerability Normalised (VUL_norm)

Definition: The Population Vulnerability score transformed to a common 0-1 scale.

Formula: $VUL_norm = (Population\ Vulnerability - PV_min) / (PV_max - PV_min)$

Interpretation: 0 = lowest relative population vulnerability; 1 = highest relative population vulnerability. VUL_norm is retained as an independent vulnerability dimension in the final hotspot model.

Access Deficit

Definition: An inverse measure of relative accessibility derived from the normalised Composite Accessibility Index.

Formula: $Access\ Deficit = 1 - CAI_norm$

Interpretation: 0 = lowest access deficit / strongest relative accessibility; 1 = highest access deficit / weakest relative accessibility. Higher values identify areas where accessibility conditions are comparatively poor.

Hotspot Score

Definition: A continuous spatial-prioritisation measure combining access deficit with population vulnerability.

Formula: $Hotspot\ Score = 0.40(Access\ Deficit) + 0.60(VUL_norm)$

Component Dimension	Assigned Model Weight (%)
Access Deficit (1 - CAI_norm)	40%
Population Vulnerability Normalised (VUL_norm)	60%

Interpretation: Higher Hotspot Scores identify areas where poor public transport accessibility coincides with higher population vulnerability. The weighting gives greater influence to population vulnerability (60%) than to physical access deficit (40%).

Intervention Priority

Definition: A categorical planning classification translating the continuous accessibility-vulnerability relationship into five intervention priority classes.

Priority Class	Recommended Planning Definition
Critical Hotspot	Very low accessibility combined with high population vulnerability
High Priority	Low accessibility combined with elevated population vulnerability
Moderate	Intermediate accessibility and population vulnerability
Low Priority	Reduced physical accessibility but comparatively lower population vulnerability
Well Served	High public transport accessibility relative to population vulnerability

Interpretation: Intervention Priority is a planning classification, not simply another continuous index. It provides an interpretable basis for identifying areas where transport interventions, service improvements or further accessibility assessment may be warranted.

3.4 Study Area Context & Census Variables

The Bandon study area encompasses the main town centre and surrounding residential developments. The demographic and socio-economic variables utilized from Census 2022 Small Area statistics are detailed in Table 5.

Census Variable Field	Description	Analytical Domain & Relevance
T1_1AGE70_74T to GE_85T	Population aged 70+	Elderly Ratio numerator; transit dependency indicator.
T1_1AGETT	Total population	Denominator across all model ratio calculations.
T4_5RP / T8_1_RT	Retired persons	Economic status domain; non-work transit demand.
T8_1_LTUT	Long-term unemployed	Socio-economic vulnerability domain.
T8_1_UTWSDT	Unable to work due to sickness/disability	Economic Vulnerability numerator.
T12_1_T	Persons with a disability	Health & Disability Vulnerability numerator.
T12_3_BT	Bad general health	Health & Disability Vulnerability numerator.
T12_3_VBT	Very bad general health	Health & Disability Vulnerability numerator.
T15_1_NC	Households without a car	No-Car Ratio numerator; car dependency proxy.
T11_1_BUT	Commuting by bus/minibus	Observed public transport usage.
T11_2_T1-T8	Departure time for work/school	Commuter travel demand timing analysis.

3.5 Master Map Series

The spatial evidence structure is organized across seven core maps (Map 01.00 through Map 01.06), each highlighting a specific structural dimension of the public transport landscape in Bandon.

Map ID	Master Title & Subtitle	Key Cartographic Message & Finding
Map 01.00	Base Context & Transit Infrastructure Bandon Built-Up Areas (BUA) and Network Baseline	Establishes spatial boundaries, arterial routes, and the distribution of active bus stops.
Map 01.01	Physical Pedestrian Catchments Network-Based Isochronal Walkability Analysis	Highlights 5, 10, and 15 minute walking catchments at 5 km/h along the road network.
Map 01.02	First/Last-Mile Infrastructure Quality Footpath Coverage and Pedestrian Crossing Networks	Converts walking catchments into Pedestrian Accessibility scores (1.0, 0.6, 0.3, 0.0).
Map 01.03	Headway & Operational Frequency Peak-Period Transit Service Quality	Maps peak-hour bus frequencies and service headways derived from GTFS data.
Map 01.04	Demographic Vulnerability & Demand Spatial Distribution of Priority Transit-Dependent Populations	Displays composite Population Vulnerability across Census Small Areas.
Map 01.05	Composite Accessibility Index (CAI) Integrated Spatial-Temporal Transit Performance	Combines Pedestrian Accessibility and service frequency into an integrated performance score.
Map 01.06	Targeted Intervention Framework Spatial Decision-Support Priorities for Capital Investment	Overlays Access Deficit (40%) and VUL_norm (60%) into Intervention Priority classes.

3.6 Data Infrastructure and Sources

The study integrates NTA GTFS schedule feeds, CSO Census 2022 Small Area statistics, OpenStreetMap pedestrian networks, and Tailte Éireann boundary data:

Data Source	Contribution to the Framework
GTFS Stops, Routes & Trips (NTA)	Public transport locations, routing, and peak timetable
OpenStreetMap	Pedestrian network routing and street topology
CSO Census 2022	Socio-demographic, health, age, and car-ownership indicators
Small Areas - Tailte Éireann / CSO	Primary spatial reporting geography

4. Results and Spatial Interpretation

The map series provides an immediate visual reading of the spatial structure of public transport accessibility in Bandon. The most consistent pattern is a stronger accessibility core around the town centre and weaker conditions toward peripheral areas.

How to read the results. The map series should be read sequentially. Maps 01.00-01.03 describe the transport network and its physical and temporal accessibility; Map 01.04 describes population vulnerability; Map 01.05 integrates pedestrian accessibility with inverse vulnerability; and Map 01.06 translates the combined evidence into intervention-priority classes. The final priority map should therefore be interpreted as a screening and decision-support layer, not as a substitute for site-level investigation.

4.1 Map 01.00 - Base Context & Transit Infrastructure

SEL - 002 01.00 | BASE CONTEXT & TRANSIT INFRASTRUCTURE

Bandon Built Up Areas (BUA) and Network Baseline



Legend

- Bus stop
- Study area
- Primary roads
- Local streets

Context Map



STUDY AREA

Bandon, West Cork, is a growing regional town with a population of 8,196 (Census 2022), located approximately 30 km southwest of Cork City. Its role as both a commuter town and gateway to West Cork makes public transport accessibility an important consideration for residents and visitors.

KEY MESSAGE

The study establishes the spatial context for assessing public transport accessibility in Bandon, showing the study boundary, public transport stops and the local road network that supports movement throughout the town.

DATA SOURCES

CSO · OpenStreetMap · NTA · Tailte Éireann

METHOD & GEOGRAPHY

Accessibility-Vulnerability Spatial Integration Model (AVSIM) · Irish Transverse Mercator (ITM)



Description

This map establishes the spatial baseline for the analysis. It locates the study boundary, the built-up settlement, the principal road network and the active public transport stops used in the subsequent accessibility assessment. It provides the reference geography for interpreting the walking-access, service-frequency, vulnerability and priority maps that follow.

Key Message

The public transport network is concentrated around the established town settlement, creating the spatial baseline for the core-periphery pattern developed through the remaining analysis. Peripheral residential areas are more spatially separated from the principal stop locations and therefore require closer examination of network access.

4.2 Map 01.01 - Physical Pedestrian Catchments

SEL - 002 01.01 | PHYSICAL PEDESTRIAN CATCHMENTS

Network-Based Isochronal Walkability Analysis



STUDY AREA

Bandon, West Cork, is a growing regional town with a population of 8,196 (Census 2022), located approximately 30 km southwest of Cork City. Its role as both a commuter town and gateway to West Cork makes public transport accessibility an important consideration for residents and visitors.

KEY MESSAGE

Network-based service areas identify areas reachable from public transport stops within 5, 10 and 15 minutes of walking, using a 5 km/h walking speed. Accessibility decreases with increasing walking time, highlighting areas located farther from public transport stops.

DATA SOURCES

CSO · OpenStreetMap · NTA · Tailte Éireann

METHOD & GEOGRAPHY

Accessibility-Vulnerability Spatial Integration Model (AVSIM) · Irish Transverse Mercator (ITM)



Description

This map measures access to public transport through the pedestrian network rather than straight-line distance. Network service areas are shown for 0-5, 5-10 and 10-15 minutes of walking at the assumed speed of 5 km/h, with locations beyond 15 minutes treated as outside the defined accessibility threshold. The result shows how the structure of the walking network influences practical access to the bus network.

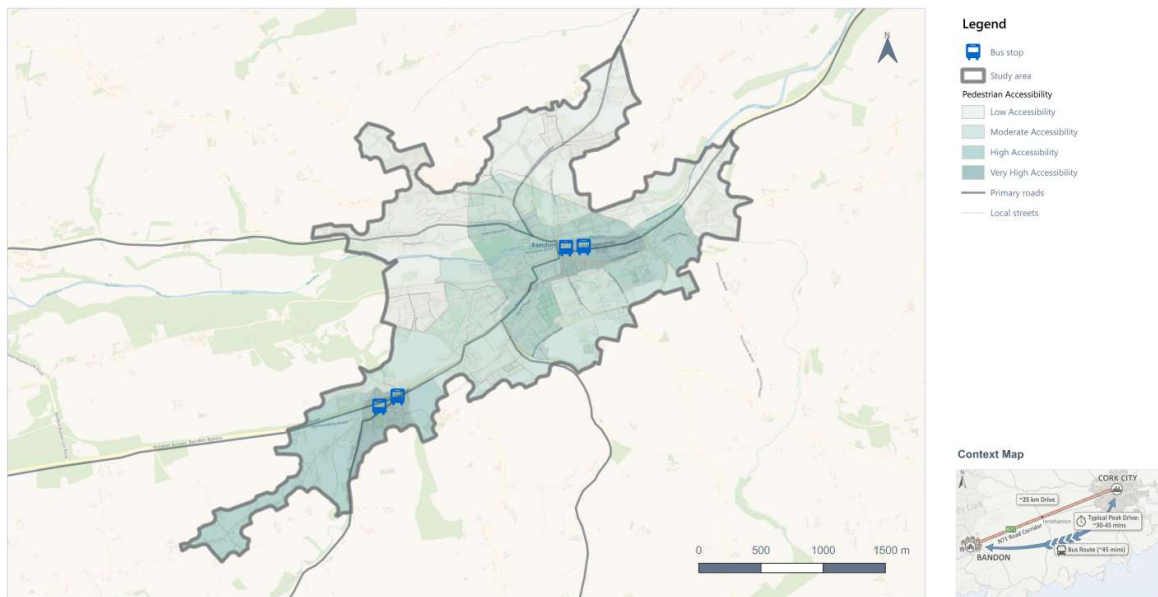
Key Message

Walking accessibility is strongest close to the principal stops and progressively weakens with increasing network walking time. The map therefore identifies a clear transition from better-connected central areas to less accessible peripheral residential areas, where reaching a stop requires longer walking journeys.

4.3 Map 01.02 - First/Last-Mile Infrastructure Quality

SEL - 002 01.02 | FIRST/LAST-MILE INFRASTRUCTURE QUALITY

Footpath Coverage and Pedestrian Crossing Networks



STUDY AREA

Bandon, West Cork, is a growing regional town with a population of 8,196 (Census 2022), located approximately 30 km southwest of Cork City. Its role as both a commuter town and gateway to West Cork makes public transport accessibility an important consideration for residents and visitors.

KEY MESSAGE

The study area shows a clear concentration of higher walking accessibility around Bandon's central area and key public transport stops, while accessibility decreases towards the outer parts of the town. This highlights the importance of improving pedestrian connections between residential areas and bus stops to support more equitable access to public transport.

DATA SOURCES

CSO · OpenStreetMap · NTA · Tailte Éireann

METHOD & GEOGRAPHY

Accessibility-Vulnerability Spatial Integration Model (AVSIM) · Irish Transverse Mercator (ITM)



Description

This map focuses on the pedestrian infrastructure that connects residents to public transport stops. Footpath coverage and pedestrian crossing networks are considered alongside the walking catchments to identify where a nominally reachable stop may be less effectively connected by the pedestrian environment. It complements Map 01.01 by moving from travel time alone to the quality and continuity of the first/last-mile connection.

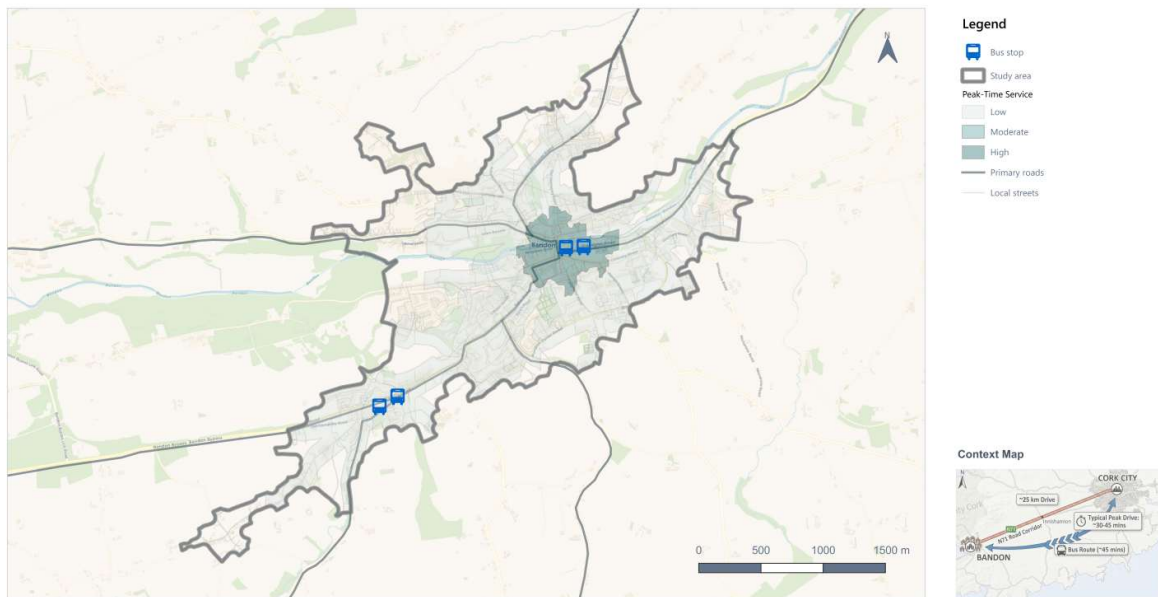
Key Message

Effective access depends not only on the distance to a stop but also on the continuity of the pedestrian route. Gaps in footpaths or crossing provision can create local barriers between residential areas and the public transport network, particularly along routes connecting peripheral neighbourhoods with central stops.

4.4 Map 01.03 - Headway & Operational Frequency

SEL - 002 01.03 | HEADWAY & OPERATIONAL FREQUENCY

Peak-Period Transit Service Quality



STUDY AREA

Bandon, West Cork, is a growing regional town with a population of 8,196 (Census 2022), located approximately 30 km southwest of Cork City. Its role as both a commuter town and gateway to West Cork makes public transport accessibility an important consideration for residents and visitors.

KEY MESSAGE

The map shows that public transport coverage is concentrated around Bandon's central urban area, with limited access in the town's outer areas. This highlights gaps in public transport reach, particularly for communities located away from the main bus routes and stops.

DATA SOURCES

CSO · OpenStreetMap · NTA · Tailte Éireann

METHOD & GEOGRAPHY

Accessibility-Vulnerability Spatial Integration Model (AVSIM) · Irish Transverse Mercator (ITM)



Description

This map adds the temporal dimension to the accessibility assessment by examining scheduled peak-period bus frequency and headway from GTFS data. Frequency indicates how often services are scheduled, while headway represents the interval between successive services. Together, these measures show where the network offers more or less frequent opportunities to travel during the assessed peak period.

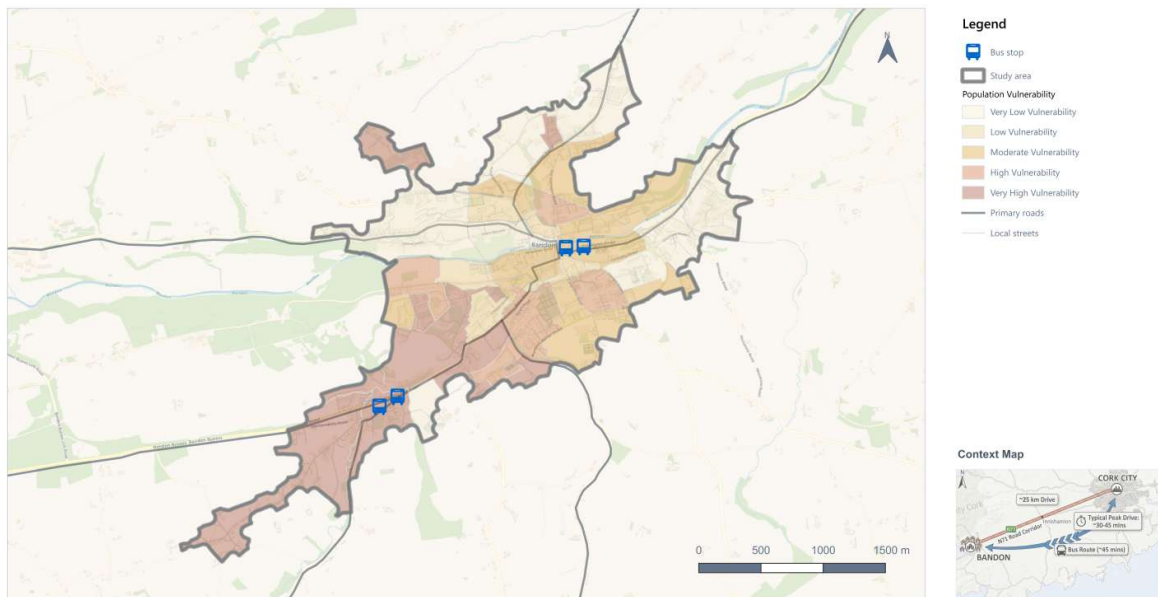
Key Message

Peak-period service availability is an important complement to physical walking access. The central corridor benefits from stronger scheduled service availability, while areas with weaker service levels may face longer waits even where a stop is within a reasonable walking distance. The frequency map should therefore be interpreted alongside the pedestrian-access results.

4.5 Map 01.04 - Demographic Vulnerability & Demand

SEL - 002 01.04 | DEMOGRAPHIC VULNERABILITY & DEMAND

Spatial Distribution of Priority Transit-Dependent Populations



STUDY AREA

Bandon, West Cork, is a growing regional town with a population of 8,196 (Census 2022), located approximately 30 km southwest of Cork City. Its role as both a commuter town and gateway to West Cork makes public transport accessibility an important consideration for residents and visitors.

KEY MESSAGE

The map highlights higher population vulnerability across several residential areas, particularly south and southwest of Bandon's centre, where vulnerability overlaps with more limited public transport accessibility. This suggests that older, health- and disability-affected, socio-economically vulnerable and transport-dependent populations may face greater barriers to accessing public transport, reinforcing the need to prioritise improved service coverage and connections in these areas.

DATA SOURCES

CSO · OpenStreetMap · NTA · Tailte Éireann

METHOD & GEOGRAPHY

Accessibility-Vulnerability Spatial Integration Model (AVSIM) · Irish Transverse Mercator (ITM)



Description

This map represents the population-need dimension of the study. Census 2022 Small Area variables are grouped into demographic vulnerability, health and mobility vulnerability, socio-economic vulnerability and transport dependence, and then combined into the Population Vulnerability measure. Higher values indicate a greater relative concentration of the characteristics included in the model; they do not represent a direct measure of individual transport use or need.

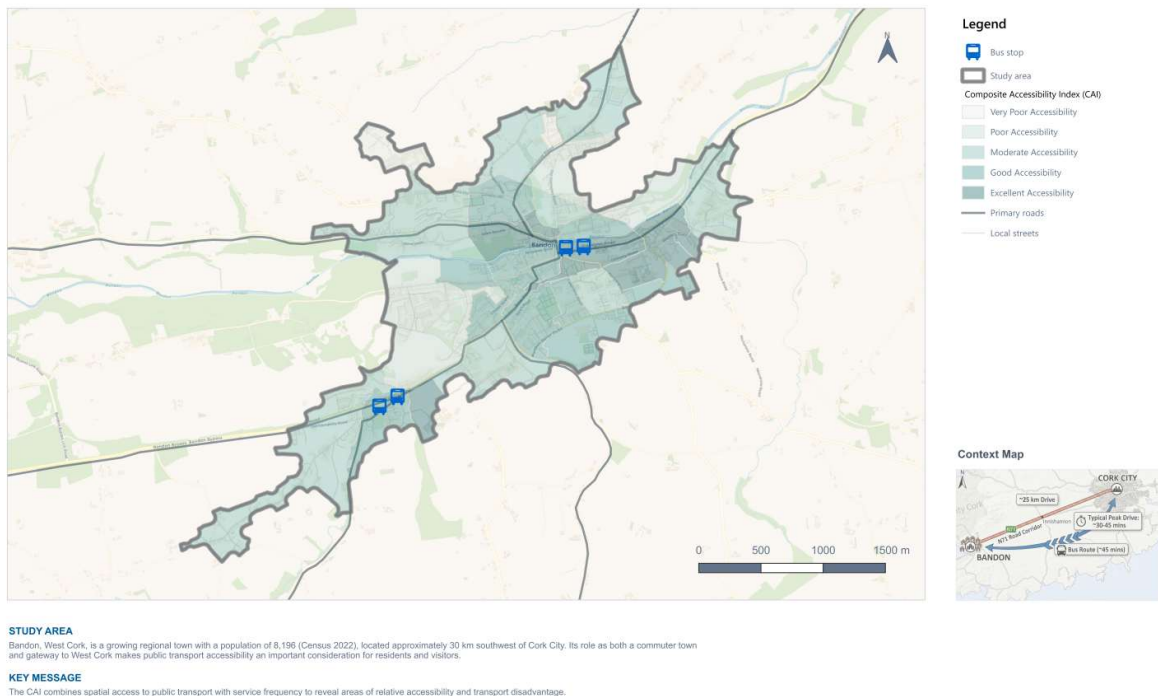
Key Message

Population vulnerability is not distributed in exactly the same way as physical public transport access. The map is therefore important for identifying places where weaker accessibility may have greater social significance, particularly where higher concentrations of older residents, health or disability indicators, economic-status vulnerability and households without a car occur.

4.6 Map 01.05 - Composite Accessibility Index (CAI)

SEL - 002 01.05 | COMPOSITE ACCESSIBILITY INDEX (CAI)

Integrated Spatial-Temporal Transit Performance



DATA SOURCES

CSO · OpenStreetMap · NTA · Tailte Éireann

METHOD & GEOGRAPHY

Accessibility-Vulnerability Spatial Integration Model (AVSIM) · Irish Transverse Mercator (ITM)



Description

The Composite Accessibility Index (CAI) combines the network-based Pedestrian Accessibility score with the inverse Population Vulnerability measure using an equal 50:50 weighting. It is therefore an integrated spatial measure of physical access considered in the context of population vulnerability. Scheduled service frequency is assessed separately in Map 01.03 and is not a direct component of the CAI formula.

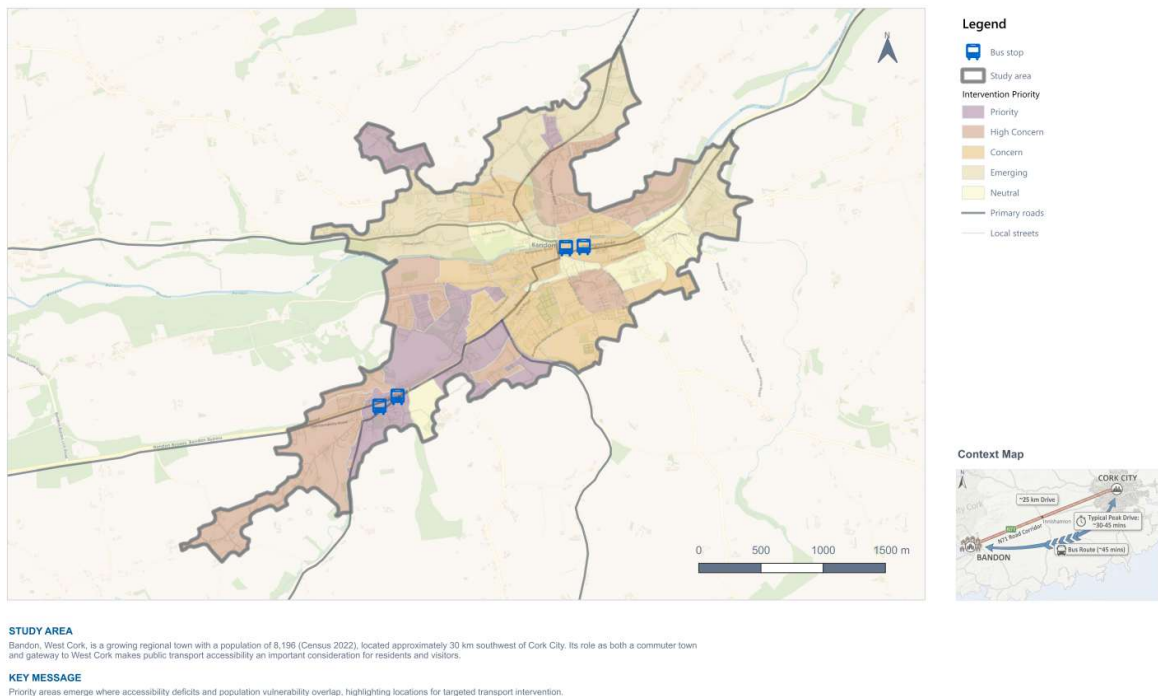
Key Message

The CAI creates a relative spatial hierarchy by distinguishing areas with stronger pedestrian access and lower relative vulnerability from areas with weaker pedestrian access and/or higher relative vulnerability. The resulting pattern reinforces the contrast between the better-connected town centre and less accessible peripheral neighbourhoods.

4.7 Map 01.06 - Targeted Intervention Framework

SEL - 002 01.06 | TARGETED INTERVENTION FRAMEWORK

Spatial Decision-Support Priorities for Capital Investment



DATA SOURCES

CSO · OpenStreetMap · NTA · Tailte Éireann

METHOD & GEOGRAPHY

Accessibility-Vulnerability Spatial Integration Model (AVSIM) · Irish Transverse Mercator (ITM)



Description

This map translates the continuous accessibility and vulnerability measures into a planning-oriented priority framework. Access Deficit contributes 40% of the Hotspot Score and normalised Population Vulnerability contributes 60%. The resulting five classes—Critical Hotspot, High Priority, Moderate, Low Priority and Well Served—identify where poor relative access coincides with higher population vulnerability.

Key Message

The highest-priority locations are concentrated in peripheral residential areas where weaker accessibility overlaps with higher relative population vulnerability. These locations should be treated as candidate areas for detailed investigation of pedestrian connections, stop provision, service arrangements or flexible transport solutions rather than as automatic prescriptions for a single intervention.

4.8 Cross-Map Synthesis

Across the seven-map sequence, the evidence progresses from the physical network baseline to walking accessibility, pedestrian infrastructure, scheduled service frequency, population vulnerability, integrated accessibility and intervention priority. The maps show a consistent spatial structure: accessibility is strongest around the town centre and declines toward several peripheral residential areas. The vulnerability analysis adds an important second dimension, showing that the places with weaker access are not necessarily the same places as those with the lowest or highest population vulnerability. The final hotspot framework is therefore more informative for planning than any individual map because it identifies where limited access and higher population vulnerability overlap.

5. Evaluation of Research Questions

The results show that public transport accessibility in Bandon is a multi-dimensional spatial issue. No single map provides the complete picture. Walking-network access establishes whether residents can physically reach the public transport system; pedestrian infrastructure indicates whether those connections are continuous and usable; GTFS frequency describes the temporal availability of scheduled services; and the Census-based vulnerability layer indicates where limited access may have greater consequences for populations with potentially greater dependence. The CAI and final hotspot framework bring these dimensions together in a transparent sequence.

- **Accessibility is More Than Distance:** Network walking time captures practical access more effectively than straight-line distance, but it does not describe service frequency or population need on its own.
- **Walking Accessibility and Urban Form:** The urban core is more closely connected to principal stops, while several peripheral residential areas fall into weaker accessibility classes.
- **Temporal Accessibility:** Peak-period GTFS frequency and headway provide a separate measure of how often scheduled services are available once a stop can be reached.
- **Population Need vs. Service Provision:** Comparing accessibility conditions with Population Vulnerability distinguishes areas where weaker access coincides with greater potential transport vulnerability.
- **Composite Accessibility Index:** The CAI combines Pedestrian Accessibility and inverse Population Vulnerability at a 50:50 weighting, creating a relative integrated measure for spatial comparison.
- **Accessibility-Vulnerability Hotspots:** The Hotspot Score gives 40% weight to Access Deficit and 60% to normalised Population Vulnerability, translating the continuous results into five planning classes.

5.1 Research Question One - Viability for Cork City Commuters

Executive Finding: Public transport is a viable alternative for residents in the central corridor, but significantly less viable for peripheral neighbourhoods:

Spatial Zone	Viability Assessment	Key Factors
Central Corridor	High	Short walking distance, high peak service frequency, direct regional bus runs to Cork City
Intermediate Areas	Moderate	Reasonable walking access, but declining peak service availability
Peripheral Areas	Limited / Low	Long walking journeys, reduced service coverage, making private cars far more competitive

5.2 Research Question Two - Identifying Vulnerable Areas and Groups

Executive Finding: Several peripheral neighbourhoods combine low public transport accessibility with higher population vulnerability, marking them as key areas of transport disadvantage.

Affected Group	Primary Barrier	Planning Need
Older Adults (70+)	Reduced walking radius to bus stops	Highly accessible, localized stop placement
Disabled / Poor Health	Physical network barriers and topography	High-quality footpaths, crossings, and low-floor buses
Long-Term Unemployed	High transport costs or poor access to job centres	Affordable, frequent peak commuting services
Car-Less Households	Total reliance on active travel and public transit	High-density transit coverage in residential sectors

5.3 Spatial Typology of Bandon

System 1 — Higher Access / Lower Relative Vulnerability: The town core and central accessibility corridor, where residents are generally closer to principal stops and the transport network is more concentrated.

System 2 — Moderate Access / Mixed Vulnerability: Transitional residential areas where access conditions vary and vulnerability is mixed, indicating a need for location-specific assessment.

System 3 — Lower Access / Higher Relative Vulnerability: Peripheral residential areas represented by Critical and High Priority classifications, where weaker access overlaps with higher relative vulnerability.

Overall results. Taken together, the analysis indicates that Bandon has a relatively well-connected central accessibility area surrounded by a more variable peripheral zone. The principal spatial issue is not simply the absence of public transport stops; it is the combination of longer walking access, pedestrian connection constraints, differences in scheduled peak-period service and the distribution of population vulnerability. The southern and southwestern peripheral neighbourhoods identified in the final synthesis warrant particular attention because weaker accessibility and higher relative vulnerability overlap there. The evidence supports targeted investigation rather than a single town-wide intervention approach.

6. Policy and Planning Implications

6.1 Priority Intervention Areas

Critical Hotspot and High Priority areas in Map 01.06 should be treated as candidate locations for detailed planning and feasibility assessment, rather than as automatic investment decisions.

6.2 Pedestrian Infrastructure Improvements

Where poor accessibility is driven primarily by walking access (Map 01.02), potential interventions include improved footpaths, safe pedestrian crossings, and removal of physical barriers along key stop access corridors.

6.3 Service Frequency Optimisation

Where walking access is reasonable but peak-time service is weaker (Map 01.03), service-frequency interventions and headway adjustments offer a better response than civil infrastructure alone.

6.4 Micro-Mobility & Demand-Responsive Transit (DRT)

Peripheral areas outside fixed-route catchments warrant assessment for Demand-Responsive Transit (DRT) or micro-mobility feeder solutions where traditional fixed-route extensions are economically inefficient.

6.5 Accessibility Monitoring

The GIS workflow can be converted into a repeatable monitoring framework, re-running metrics as GTFS schedules, population figures, and pedestrian infrastructure evolve.

6.6 Recommendations for Cork County Council and NTA

1. Targeted Spatial Assessment: Use Map 01.06 as a decision-support layer to identify candidate areas for more detailed assessment of active-travel, public transport and flexible-transit interventions.
2. Active Travel Infrastructure: Investigate footpaths, pedestrian crossings and street lighting along access corridors connecting southern and southwestern peripheral estates, including Gully, Cloughmacsimon and Kilbrogan, to central bus stops.
3. Service Frequency Assessment: Review peak timetable headways and scheduled bus frequency on Bandon-Cork City services, particularly where reasonable walking access is not matched by strong peak-period service availability.
4. Flexible Transit Assessment: Examine the feasibility of Demand-Responsive Transit (DRT) or local feeder services for low-density peripheral neighbourhoods outside or weakly served by fixed-route catchments.

7. Limitations and Future Research

7.1 Limitations

- **Walking Assumptions:** Network routing assumes a standard 5 km/h walking speed and complete street topology.
- **GTFS Limitations:** GTFS reflects scheduled timetable data rather than real-time vehicle locations or operational delays.
- **Census Data:** Population indicators rely on static Census 2022 Small Area boundaries.
- **Demand Dynamics:** Accessibility measures potential supply and spatial reach rather than observed personal transit trip demand.

7.2 Future Research Priorities

GTFS - Real-time Accessibility: Integrate GTFS - Realtime data to evaluate reliability and delay impacts.

Temporal Expansion: Evaluate off-peak, evening, and weekend accessibility patterns.

Accessibility for Schools and Healthcare: Measure accessibility to specific essential destinations and examine whether vulnerable populations have adequate access to them.

Accessibility for Active Travel: Integrate walking and cycling infrastructure quality, safety and connectivity into the accessibility framework.

Integration into Ireland Quality of Life Index: Develop transport accessibility as a spatial quality-of-life dimension that can be integrated with the wider Ireland Small Area Quality of Life Index framework.

Regional Replication: Replicate workflow across comparable Irish regional towns (e.g. Clonakilty, Skibbereen, Bantry, Midleton and Mallow).

8. Conclusion

The Bandon Public Transport Accessibility Analysis demonstrates how a reproducible GIS workflow can bring together pedestrian-network access, scheduled public transport service information and Census-based population vulnerability to explain spatial differences in transport accessibility. The principal result is a clear core-periphery structure: central areas generally have stronger access, while several southern and southwestern peripheral neighbourhoods show weaker conditions.

8.1 Summary of Findings

Public transport accessibility across Bandon is strongly centralised around the town centre. The pedestrian catchment analysis shows that access weakens as walking time to principal stops increases; the infrastructure and service-frequency maps add evidence on the quality and temporal availability of those connections. The vulnerability analysis shows that population need is spatially uneven, and the final hotspot framework identifies peripheral locations where lower relative accessibility overlaps with higher relative vulnerability. These results provide a structured basis for deciding where further pedestrian, service or flexible-transport investigations should be concentrated.

8.2 Research Contribution

The principal contribution is methodological: the study demonstrates a reproducible spatial workflow for assessing public transport accessibility in a smaller Irish town using openly available spatial data, GTFS schedules and Census 2022 Small Area information. It also provides a framework for linking transport accessibility to wider social-equity and quality-of-life analysis.

8.3 Final Reflection

The study demonstrates the value of moving beyond simple distance measures. For Bandon, transport accessibility is best understood as the interaction between where people live, how easily they can reach the network, how frequently scheduled services operate, and the characteristics of the population exposed to weaker access. The resulting framework is intended to support evidence-led prioritisation while recognising that individual interventions require further engineering, operational, demand and feasibility assessment.

References

- Accessibility Theory: Hansen (1959); Geurs & van Wee (2004)
- Transport Equity: Lucas (2012); Martens (2016)
- GIS Accessibility: ESRI Network Analyst Literature & OpenStreetMap Routing Protocols
- Irish Transport Policy: National Transport Authority (NTA); Department of Transport
- Census & Spatial Analysis: Central Statistics Office (CSO) Census 2022

Appendix A. Data Sources and Reproducibility Register

This appendix records the principal datasets represented in the current study. Exact dataset versions and extraction dates should be added when the production metadata is finalised.

Dataset	Version / Date	Purpose
CSO Census 2022 Small Areas	2022	Population and socio-demographic indicators.
Bus Éireann / NTA GTFS	GTFS feed 2026	Stops, routes, and stop times.
OpenStreetMap	Extract 2026	Walking street network and local spatial context.
Administrative / Small Area Boundaries	Tailte Éireann / CSO	Study and reporting geography.
Spatial Evidence Lab PostGIS Database	Version 1.0	Processed accessibility and vulnerability layers.

Appendix B. Map Production Register

The following register is the authoritative map naming structure for the SEL-002 deliverables.

Map	Title	Subtitle	File Name
01.00	Base Context & Transit Infrastructure	Bandon Built-Up Areas (BUA) and Network Baseline	SEL-002_01.00
01.01	Physical Pedestrian Catchments	Network-Based Isochronal Walkability Analysis	SEL-002_01.01
01.02	First/Last-Mile Infrastructure Quality	Footpath Coverage and Pedestrian Crossing Networks	SEL-002_01.02
01.03	Headway & Operational Frequency	Peak-Period Transit Service Quality	SEL-002_01.03
01.04	Demographic Vulnerability & Demand	Spatial Distribution of Priority Transit-Dependent Populations	SEL-002_01.04
01.05	Composite Accessibility Index (CAI)	Integrated Spatial-Temporal Transit Performance	SEL-002_01.05
01.06	Targeted Intervention Framework	Spatial Decision-Support Priorities for Capital Investment	SEL-002_01.06

Document Control

Project: SEL-002 - Bandon Public Transport Accessibility Analysis

Status: Final Revised Report (Updated Master Map Titles & Strategic Synthesis)

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